MIL-G-60624(MU)
9 January 1967
SUPERSEDING
FA-PD-MI-2564 Rev 0
28 September 1962

MILITARY SPECIFICATION
GENERATOR, GAS PRESSURE, PROPELLANT ACTUATED, XM17, ASSEMBLY

1. SCOPE

1.1 This specification covers Generator, Gas Pressure, Propellant Actuated, XM17, Assembly loaded with Cartridge, Aircraft Ejection Seat, Catapult, M37, for use in conjunction with other propellant actuated devices in the recovery system of training bombs.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550 - Ammunition and Special Weapons, General Specification for

STANDARDS

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1167 - Ammunition Data Card

DRAWINGS

Munitions Command

F10522490 - Generator, Gas Pressure, Propellant Actuated, XM17 Assembly.
C8864324 - Box, Packing, Ammunition for Generator, Gas Pressure, Propellant Actuated, XM17 in Fiber Container M with Cartridge (See 5.1).

FSC 1377

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C8864325 - Container, Ammunition, Fiber M for Generator, Gas Pressure, Propellant Actuated, XM17, with Cartridge (See 5.1).
IEL10522490- Inspection Equipment List for Generator, Gas Pressure, Propellant Activated, XM17, Assembly.

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Breakaway Torque.- The chamber body and cap of the assembled generator shall withstand a breakaway torque application of 400 inch pounds without loosening or deformation.

3.3 Residual Magnetism.- The generator shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.4 Functional.

3.4.1 Ballistics.- Generators loaded with accepted cartridges and temperature conditioned across the range of -65°F to 160°F shall:

3.4.1.1 Shear a 7200-pound shear joint within .150 seconds after initiation of the M37 Cartridge.

3.4.1.2 Accelerate an 80-pound weight, in a horizontal to a 10° below horizontal direction, to a minimum velocity of 120 feet per second within a maximum distance of 42 inches.

3.4.2 The generator shall not fail to fire.

3.4.3 The generator shall function without mechanical failure (See 6.1).

3.5 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

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3.5.1 Metal Defects.- All components shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.5.2 Burr.- No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.5.3 Foreign Matter.- No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance Terms and Definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 Inspection Provisions.

4.2.1 Lot.

4.2.1.1 Submission of Product.- The product shall be submitted in accordance with MIL-STD-105.

4.2.1.2 Size of lot.- A lot shall consist of an identifiable group of assembled units containing metal parts assemblies produced by a homogeneous process and loaded with accepted cartridges.

4.2.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Table I. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance

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4.2.2 (Cont'd)

number of Table I. All non-conforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE I

Lot Size	Sample Size	Major Accept	Major Reject	Minor Accept	Minor Reject
1 - 300	100% Examination	-	-	-	-
301 - 800	50	0	1	1	2
801 - 1800	90	1	2	2	3
1801 - 10,000	200	2	3	3	4

4.2.2.1 Cartridge, M37, Drawing D8593745, covering a detail of Dwg. F10522490.

	Defect	Method of Inspection
CRITICAL		
1.	Mixed cartridges	Visual
2.	Metal defective	Visual
3.	"O" ring missing	Visual
4.	Sealing disc missing, damaged or improperly secured	Visual
5.	Marking missing, incorrect or illegible	Visual
MAJOR		
101.	"O" ring split, cut or cracked	Visual
102.	Foreign matter	Visual
MINOR		
201.	Case scratched or dented	Visual
202.	Varnish missing or inadequate	Visual

4.2.2.2 Assembly, Drawing F10522490, consisting of D8593745, A10522494, B10522496 and MS-29513-20.

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4.2.2.2(Cont'd)

Defect	Method of Inspection
CRITICAL	
1. Any part missing	Visual
2. Improper assembly	Visual
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

4.2.2.3 Assembly, Drawing D10522490, consisting of A10520494, F10522493, C10522494, B10522495 and MS-29513-12.

CRITICAL	
1. Any part missing	Visual
2. Improper assembly	Visual
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

4.2.2.4 Assembly, Drawing F10522490.

CRITICAL	
1. Cartridge head not properly seated in cap	Manual
2. Assembly loose	Manual
3. "O" ring missing	Visual
4. Improper assembly	Visual
5. Marking missing, incorrect or illegible	Visual

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4.2.2.4(Cont'd)

Defect	Method of Inspection
MAJOR	
101. Total length	Gage
MINOR	
201. Shipping cap missing	Visual
202. Compound missing from setscrew	Visual
203. Location of labels	Visual
204. Varnish missing or inadequate	Visual
205. Protective finish inadequate or defective	Visual

4.2.2.5 Fiber Container, Drawing C8864325.

CRITICAL

None defined

MAJOR

None defined

MINOR

201. Gap between body and cover, max.	Manual
202. Contents loose	Manual
203. Metal ends loose or distorted	Manual
204. Cut or damaged through all impregnated layers	Visual
205. Marking missing, incorrect or illegible	Visual
206. Tape incomplete or badly wrinkled	Visual
207. Length of tear tab, min.	Visual
208. Color of tape, incorrect	Visual

4.2.2.6 Box, Packing, Drawing C8864324.

CRITICAL

None defined

MAJOR

None defined

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4.2.2.6(Cont'd)

Defect	Method of Inspection
MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Manual and Visual
203. Strapping missing, loose, broken or improperly assembled	Manual and Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking, missing, incorrect or illegible	Visual
207. Number of units, incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.2.3 Inspection.- The following tests shall be performed in accordance with 4.3.

4.2.3.1 Breakaway Torque.- Each body and cap assembly shall be subjected to a breakaway torque. Failure of any assembly to comply with the requirements of 3.2 shall be cause for rejection of the assembly.

4.2.3.2 Residual Magnetism.- Each generator shall be subjected to the test for excessive residual magnetism. Failure of an generator to comply with the requirements of 3.3 shall be cause for rejection of the generator.

4.2.3.3 Functional Tests.- The following tests will be performed by the Government.

4.2.3.3.1 Ballistics.- Thirty units shall be tested in accordance with Table II to determine compliance with the requirements of 3.4.1 through 3.4.3. The units shall be temperature conditioned as provided in 4.3.3.1. Failure of any unit to comply with the requirements shall be cause for rejection of the lot.

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TABLE II

Temperature	Sample Size
-65° to -70°F	10
65° to 75°F	10
160° to 165°F	10

4.2.4 Inspection Equipment.- The examination and tests shall be made using equipment listed on IEL-10522490.

4.3 Test Methods and Procedures.

4.3.1 Breakaway Torque.- A torque wrench, set at the torque requirements specified in 3.2 shall be used in conjunction with an adaptor to determine the adequacy of the joint. The generator body shall be secured in a holding device and the breakaway torque shall be applied to cap.

4.3.2 Residual Magnetism.- The generator shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the generator 6 inches from and at the same height as the compass in a north-south horizontal position (cap end at South) with the compass set in an east-west heading as shown in Figure I. The generator shall be moved slowly past the compass. This procedure shall be repeated for each 90 degree rotation of the generator.

4.3.3 Functional Tests.

4.3.3.1 Temperature Conditioning.- Prior to conducting the Ballistic test specified in 4.3.3.3, the generator samples shall be subjected to the temperature conditions specified in 3.4.1. A simulated generator loaded with a dummy M37 cartridge containing a temperature sensitive element shall be placed in the temperature conditioning chamber at the same time as the generator samples. The conditioning time shall be a minimum of one hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

4.3.3.1.1 Upon completion of the required temperature conditioning, each generator sample shall be ballistically tested within 5 minutes after removal from the conditioning chamber. Samples shall be reconditioned if this time is exceeded. When reconditioning generators at minus 65 degrees F, all condensation shall be removed from the interior surfaces before

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returning the generator to the conditioning chamber.

4.3.3.2 Method of testing.- The generator tested under the conditions as specified herein shall be installed in the Test Fixture listed in IEL-10522490. The fixture shall be set up so that the test weight will be propelled in the direction specified in 3.4.1.2 when the generator is actuated. A velocity and acceleration recording system shall be used to determine velocity and acceleration of the moving test weight in the ballistic testing of any generator under load conditions.

4.3.3.3 Ballistic Test.- Utilize the test fixture specified in 4.3.3.2. The weight assembly shall be installed in the fixture and secured with a shear ring of 7200 pound total load capacity. An XM86 initiator or simulated method shall be used to actuate the generator under test. A pressure sensitive device shall be used to measure the pressure exerted to actuate the generator under test. A pressure sensitive device shall be used to measure the pressure exerted to actuate the generator and the amount of pressure expended by the generator when fired. The pressure sensitive device shall be inserted into the system by means of a Tee coupling between the generator and the actuating device. The generator sample shall be tested in accordance with the functional requirements of 3.4 within the time specified and the results recorded.

5. PREPARATION FOR DELIVERY

5.1 Packing and Marking.- Level A - Packing and Marking shall be in accordance with Drawings C8864324 and C8864325.

5.2 Data Cards.- Data Card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

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PILOT's STANDBY COMPASS, TYPE AN-5766



PLAN VIEW

Note.- EJECTOR AND COMPASS SHALL LIE IN A HORIZONTAL PLANE

LAYOUT FOR DETERMINATION OF RESIDUAL MAGNETISM IN GENERATOR, GAS PRESSURE, PROPELLANT ACTUATED, XM17

Figure I

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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